

Shubham Kumar Sharma

Ajmer, RJ | shubhamsharma86900@gmail.com | +91-8690097812 | Portfolio | LinkedIn | GitHub

Summary

Software Engineer with deployed systems across full-stack web and AWS cloud infrastructure. Built a live DSA visualizer in **React/TypeScript** (Node.js + Express backend) and a **RAG** pipeline on **AWS Bedrock** with Terraform-provisioned infra — both in production. **Top 1,500 of 50,000+** AWS AI & ML Scholars (Udacity & AWS). Java · TypeScript · Python · React · Spring Boot · Node.js · Docker · Terraform.

Technical Skills

Languages: Java, TypeScript, Python, C, SQL	Frontend: React, React Native, Tailwind CSS, Vite
Backend: Spring Boot, Node.js, Express, Prisma, Docker	Cloud / AI: AWS (SageMaker, Bedrock), Terraform, IBM Cloud
Data: PostgreSQL, MySQL, Supabase, Aurora Serverless	Concepts: RAG, GenAI, Microservices, REST APIs, Git

Experience

IBM SkillsBuild Intern — AI & Cloud Technologies Edunet Foundation · Remote	July 2025 – Aug 2025
- Trained and deployed a Snap Random Forest classifier on IBM Watsonx.ai — 95.8% accuracy, outperforming logistic regression baseline by ~12pp on a government benefit applicant dataset. - Designed a decision support tool cutting manual applicant review time by 70% — removing the human classification bottleneck from a government benefit distribution pipeline.	

Projects

Intelligent Document Querying System (AWS Bedrock · RAG · Terraform · Production)	GitHub
Stack: AWS Bedrock, Terraform, Aurora Serverless PostgreSQL, S3, Python, Streamlit. - Built an end-to-end RAG pipeline — AWS Bedrock Knowledge Bases, Aurora Serverless PostgreSQL, S3 — delivering sub-second natural language search over multi-page technical PDFs. - Provisioned full AWS infrastructure via Terraform across two modular stacks (VPC, NAT, Aurora Serverless, S3, Bedrock KB, IAM) — reproducible end-to-end from a single terraform apply. - Implemented prompt validation via a Bedrock model pre-check — filtering irrelevant queries before KB invocation — and tuned temperature and top_p to balance retrieval precision with response diversity.	
Data Structures & Algorithms Visualizer (Live · 60 FPS · Full-Stack · 20+ Algorithms)	Live GitHub
Stack: React, TypeScript, Node.js, Express, Prisma, PostgreSQL, Zustand, Tailwind CSS, SVG, Canvas. - Built a custom Animation Engine separating algorithmic logic from UI rendering — stable 60 FPS on O(n²) sorts across 100+ element arrays, zero frame drops. - Replaced React state with Zustand — 30% fewer re-renders on worst-case O(n log n) sorts, verified via React DevTools Profiler; rendered 6+ data structures with live complexity overlays at every step. - Architected a full-stack system with Node.js + Express + Prisma + PostgreSQL backend, Jest test suite (health, catalog, auth flows), and GitHub Actions CI/CD — supporting 20+ algorithms across Arrays, Trees, Graphs (Dijkstra, A*, Kruskal, Prim), and Sorting.	
Quantum Circuit Simulator (Qiskit · 5-Layer Architecture · IBM Quantum)	GitHub
Stack: Python, Qiskit, Matplotlib, NumPy, Tkinter. - Architected a 5-layer production-grade simulator (Presentation → Application → Quantum Abstraction → Execution → Infrastructure) — domain logic decoupled from execution engine via Dependency Inversion, with adapters for Qiskit Aer and IBM Quantum hardware. - Implemented 15+ gate types (H, CNOT, S, T, RX, RY, RZ, SWAP) with algorithm presets for Bell State, GHZ State, and Teleportation; dual CLI + Tkinter GUI with embedded Bloch Sphere, state vector, and histogram visualizations. - Applied SOLID principles with layer-wise unit tests and a classical comparison module — enabling side-by-side quantum vs. deterministic analysis; multi-qubit state computation optimized via NumPy vectorization.	

Education

Engineering College, Ajmer — B.Tech in Information Technology CGPA: 9.06/10.0	Sept 2023 – Sept 2027
AWS AI & ML Scholars — Future AWS AI Engineer Nanodegree Udacity & AWS	Aug – Nov 2025
Selected: Top 1,500 of 50,000+ global applicants · ML, GenAI, SageMaker, Bedrock, Terraform	

Certifications

AWS ML Foundations — AWS (2025)	Intro to GenAI with AWS — Udacity (2025)	Getting Started with AI — IBM (2025)
Python Essentials 1 & 2 — Cisco (2025)		